NEWTON SCHOOL

PRE-ALGEBRA

Area of Shapes Worksheet — 20 Questions

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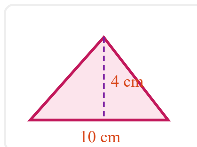
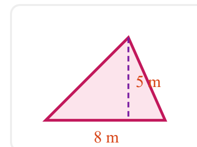
Quick formulas: Rectangle/parallelogram: $b \cdot h$ · Triangle: $\dfrac{b \cdot h}{2}$ · Trapezoid: $\dfrac{(b_1 + b_2) \cdot h}{2}$

Part A — Triangles (use perpendicular height)

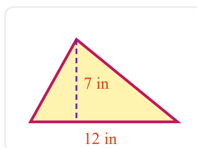
1) Triangle, b = 10 cm, h = 4 cm. Area = ____________



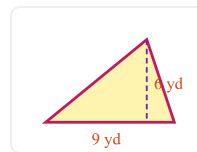
2) Triangle, b = 8 m, h = 5 m. Area = ____________



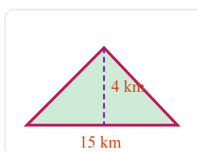
3) Triangle, b = 12 in, h = 7 in. Area = ____________



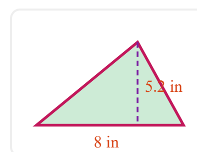
4) Triangle, b = 9 yd, h = 6 yd. Area = ____________



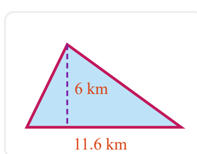
5) Triangle, b = 15 km, h = 4 km. Area = ____________



6) Triangle, b = 8 in, h = 5.2 in (slant 6 in given but not needed). Area = ____________

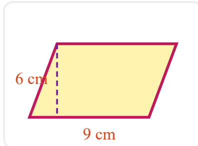


7) Triangle, b = 11.6 km, h = 6 km. Area = ____________

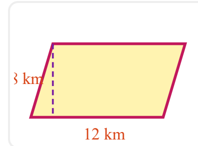


Part B — Parallelograms

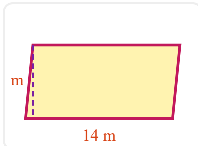
8) Parallelogram, b = 9 cm, h = 6 cm. Area = ____________



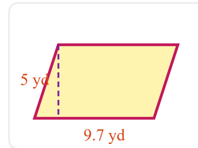
9) Parallelogram, b = 12 km, h = 8 km. Area = ____________



10) Parallelogram, b = 14 m, h = 5 m. Area = ____________

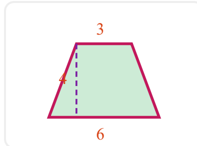


11) Parallelogram, b = 9.7 yd, h = 5 yd (slant side 5.8 yd is extra info). Area = ____________

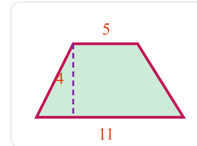


Part C — Trapezoids (use either Way A or Way B from the Notes!)

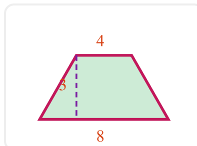
12) Trapezoid, b_1 = 3 cm, b_2 = 6 cm, h = 4 cm. Area = ____________



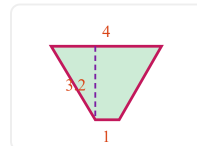
13) Trapezoid, b_1 = 5 m, b_2 = 11 m, h = 4 m. Area = ____________



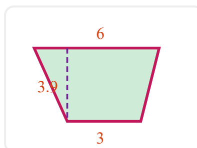
14) Trapezoid, b_1 = 4 km, b_2 = 8 km, h = 3 km. Area = ____________



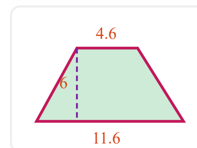
15) Trapezoid, b_1 = 4 yd, b_2 = 1 yd, h = 3.2 yd. Area = ____________



16) Trapezoid, b_1 = 6 m, b_2 = 3 m, h = 3.9 m. Area = ____________



17) Trapezoid, b_1 = 4.6 km, b_2 = 11.6 km, h = 6 km. Area = ____________



Part D — Find the missing dimension (run the formula backwards!)

18) A triangle has area **24 cm²** and height **6 cm**. Find the base. b = ____________

19) A parallelogram has area **72 m²** and base **9 m**. Find the height. h = ____________

20) A trapezoid has area **96 cm²**, height **8 cm**, and one base $b_1 =$ **10 cm**. Find the other base. $b_2 =$ ____________